

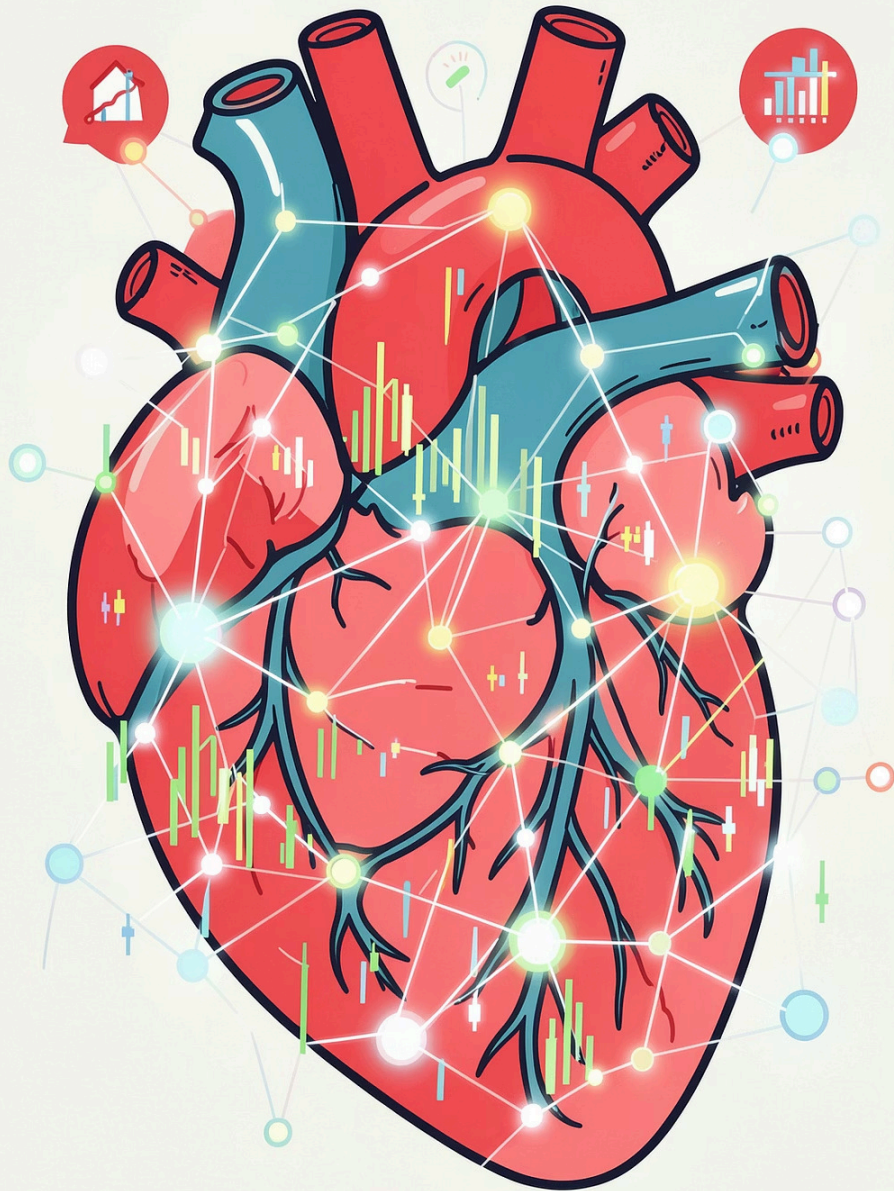
Heart Disease Prediction Using Machine Learning

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Department: **Data Science and Data Analysis**

Technology Used: **Python, Machine Learning, Streamlit**



Heart Disease Prediction Using Machine Learning

Heart disease is one of the leading causes of death worldwide. Machine Learning helps analyze medical data and predict heart disease risk at an early stage — enabling faster, smarter, and more accurate clinical decisions.



Early Diagnosis

Detect risk before symptoms escalate



Faster Decisions

Accelerate clinical decision-making



Healthcare Support

Improved support for medical professionals



Reduced Risk

Lower medical risk through proactive care

Aim and Objectives

Our Aim

To compare multiple Machine Learning algorithms and identify the most effective model for heart disease prediction.

Key Objectives

01

Analyze the Heart Disease Dataset

02

Perform data preprocessing and visualization

03

Implement Logistic Regression, KNN, and Decision Tree

04

Compare Accuracy and F1-Score

05

Select the best-performing model

06

Develop a Streamlit web application

Dataset Description

KAGGLE HEART DISEASE DATASET

Features Used

Age & Gender

Demographic indicators of patient profile

Chest Pain

Type and severity of chest pain reported

Blood Pressure

Resting blood pressure measurement

Cholesterol

Serum cholesterol level in mg/dl

Heart Rate

Maximum heart rate achieved

Exercise Chest Pain

Chest pain induced by exercise

Target Variable

```
abc.py > ...
abc.py > ...
1 import pandas as pd
2 df=pd.read_csv("heart.csv")
3 print(df)
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	52	1	0	125	212	0	1	168	0	1.0	2	2	3	0
1	53	1	0	140	203	1	0	155	1	3.1	0	0	3	0
2	70	1	0	145	174	0	1	125	1	2.6	0	0	3	0
3	61	1	0	148	203	0	1	161	0	0.0	2	1	3	0
4	62	0	0	138	294	1	1	106	0	1.9	1	3	2	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
1020	59	1	1	140	221	0	1	164	1	0.0	2	0	2	1
1021	60	1	0	125	258	0	0	141	1	2.8	1	1	3	0
1022	47	1	0	110	275	0	0	118	1	1.0	1	1	2	0
1023	50	0	0	110	254	0	0	159	0	0.0	2	0	2	1
1024	54	1	0	120	188	0	1	113	0	1.4	1	1	3	0

[1025 rows x 14 columns]

PS F:\Python>

i The target variable is binary — making this a classic supervised classification problem suitable for Logistic Regression, KNN, and Decision Tree models.

Data Preprocessing



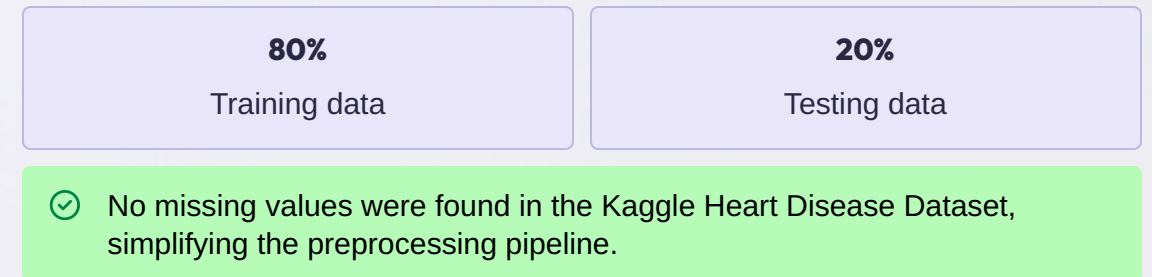
Each step ensures the data is clean, relevant, and properly divided before feeding into machine learning models.

Train-Test Split Code

```
import pandas as pd
df = pd.read_csv("heart.csv")

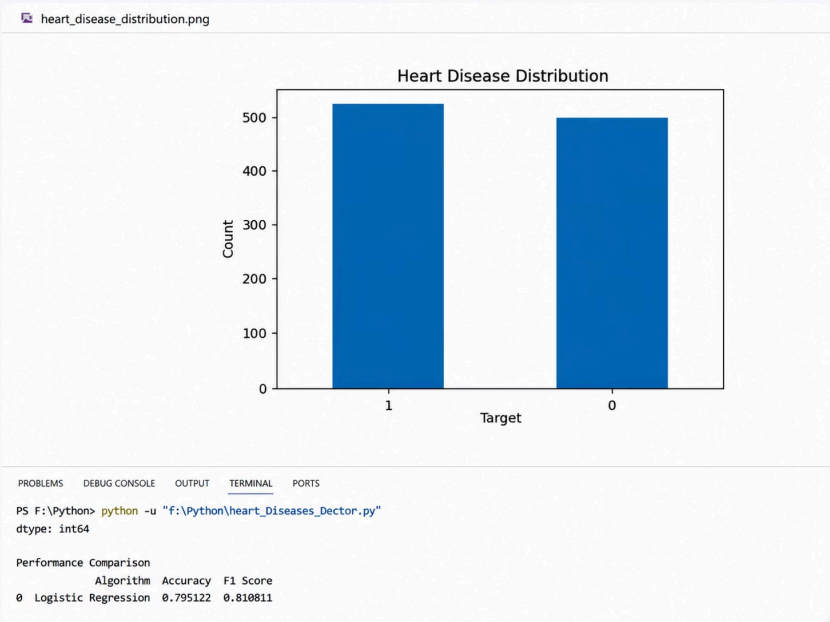
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
```

Preprocessing Summary

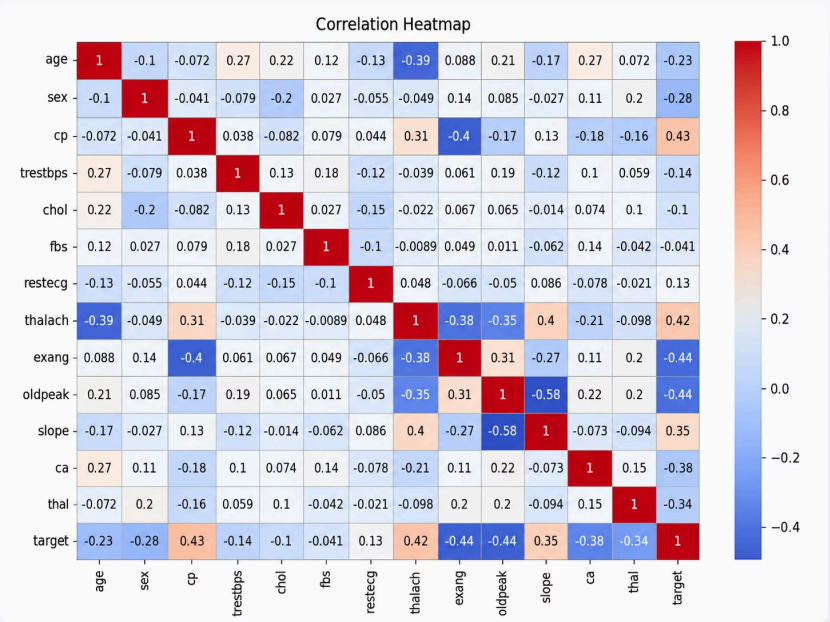


Exploratory Data Analysis

Performed Exploratory Data Analysis to understand data patterns and feature relationships within the Heart Disease Dataset.



heart disease distribution



correction heatmap

Observation: Features such as **Chest Pain**, **Cholesterol**, and **Heart Rate** show significant impact on prediction outcomes and are among the strongest correlating variables in the dataset.

Logistic Regression

Logistic Regression is used for binary classification problems — making it a natural fit for predicting the presence or absence of heart disease.

Advantages



Simple & Efficient

Minimal computational overhead with strong baseline performance



Fast Training

Converges quickly even on larger datasets



Easy Interpretation

Coefficients directly indicate feature importance

Implementation

```
from sklearn.linear_model import LogisticRegression
```

```
lr = LogisticRegression()  
lr.fit(X_train, y_train)
```

```
y_pred = lr.predict(X_test)
```

K-Nearest Neighbors (KNN)

The KNN algorithm classifies data based on the nearest neighboring records in the feature space — a non-parametric, instance-based learning approach.

Advantages

Easy to Implement

No explicit training phase required — the model memorizes the training data directly

Effective for Small Datasets

Performs well when the dataset is compact and well-structured like the Heart Disease Dataset

Implementation (k=5)

```
from sklearn.neighbors import KNeighborsClassifier

knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)

y_pred = knn.predict(X_test)
```

 The value **n_neighbors=5** means the algorithm considers the 5 closest data points when making a classification decision.

Decision Tree

The Decision Tree algorithm uses a series of decision rules derived from the training data to classify patient records into heart disease or no heart disease categories.

Advantages



Easy Visualization

Tree structure can be plotted and inspected directly



High Interpretability

Decision rules are human-readable and explainable



Good Prediction

Strong performance on structured tabular data

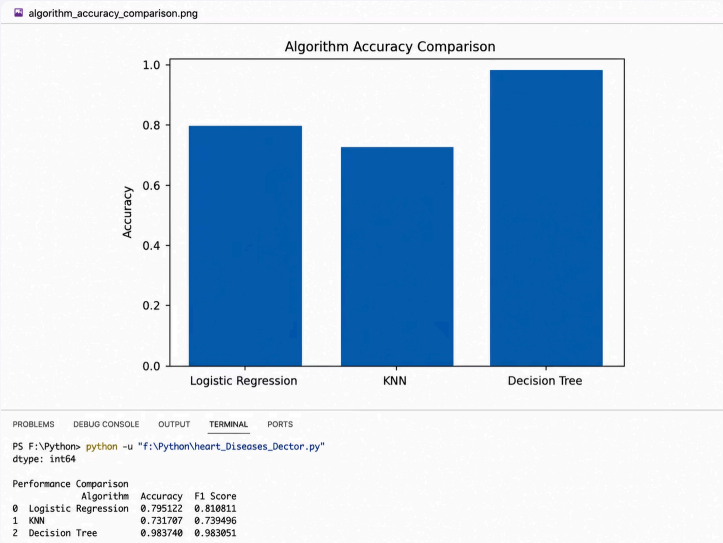
Implementation

```
from sklearn.tree import DecisionTreeClassifier

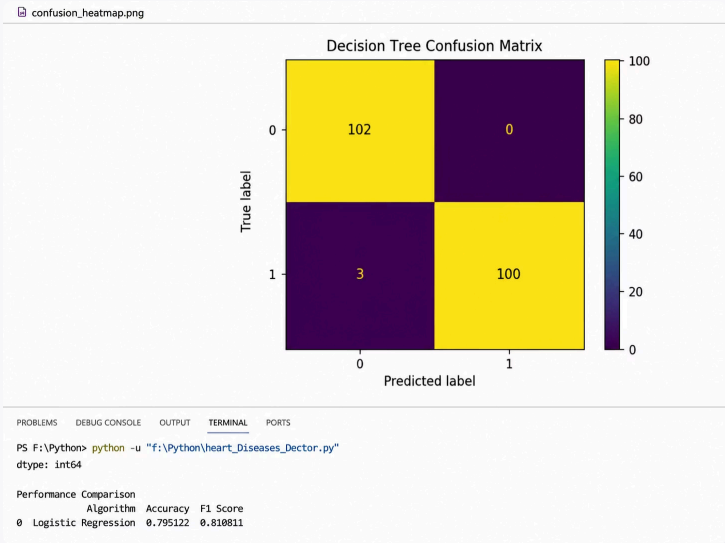
dt = DecisionTreeClassifier()
dt.fit(X_train, y_train)

y_pred = dt.predict(X_test)
```

Performance Comparison



Accuracy Comparison of Algorithms



desicion time Confusion Matrix

Results Summary

Algorithm	Accuracy	F1 Score
Logistic Regression	0.795122	0.810811
K-Nearest Neighbors (KNN)	0.731707	0.734300
Decision Tree	0.985366	0.985222

✔ **Best Model:** The algorithm achieving the highest combined Accuracy and F1 Score will be selected as the final model for deployment in the Streamlit web application.

Streamlit Web Application

Heart Disease Prediction System

Developed a Streamlit-based web application for real-time heart disease prediction — enabling clinicians and users to input patient data and receive instant risk assessments.

 **Heart Disease Prediction System**

Age

29.00

-

+

Sex

Female

▼

Chest Pain Type (0-3)

0.00

-

+

Resting Blood Pressure

110.00

-

+

Cholesterol

170.00

-

+

2.00

-

+

Number of Major Vessels (0-4)

2.00

-

+

Thal (0-3)

2.00

-

+

Predict

Prediction Generated Successfully

✔ No Heart Disease Detected

User-Friendly Interface

Clean, intuitive form for entering patient data



Real-Time Prediction

Instant results powered by the trained ML model



Technologies Used

Streamlit · Python · Scikit-learn ML Model

Conclusion and Future Scope

Our project successfully demonstrated the application of machine learning in predicting heart disease, laying a strong foundation for future advancements.

Key Accomplishments

- Analyzed the Heart Disease Dataset comprehensively
- Implemented Logistic Regression, KNN, and Decision Tree models
- Compared model performance using Accuracy and F1 Score
- Developed a functional Streamlit web application
- Selected the best-performing model for deployment

Future Scope & Directions



Deep Learning Models

Explore advanced neural networks for improved accuracy



Mobile Application

Develop a mobile app for convenient patient risk assessment



Hospital Integration

Integrate the model into existing clinical systems



Real-Time Monitoring

Enable continuous patient data analysis and alerts

THANK YOU